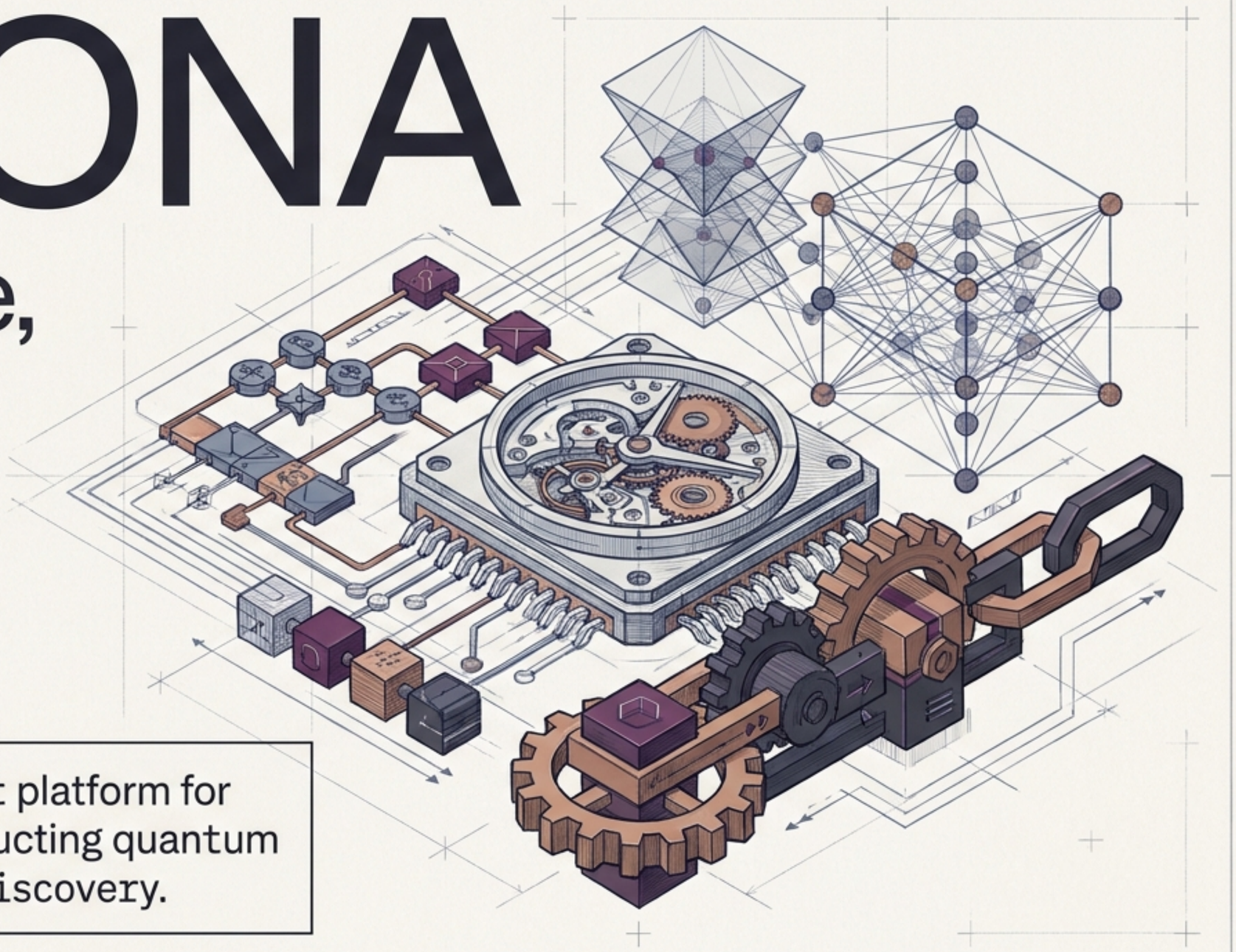
KUBONA

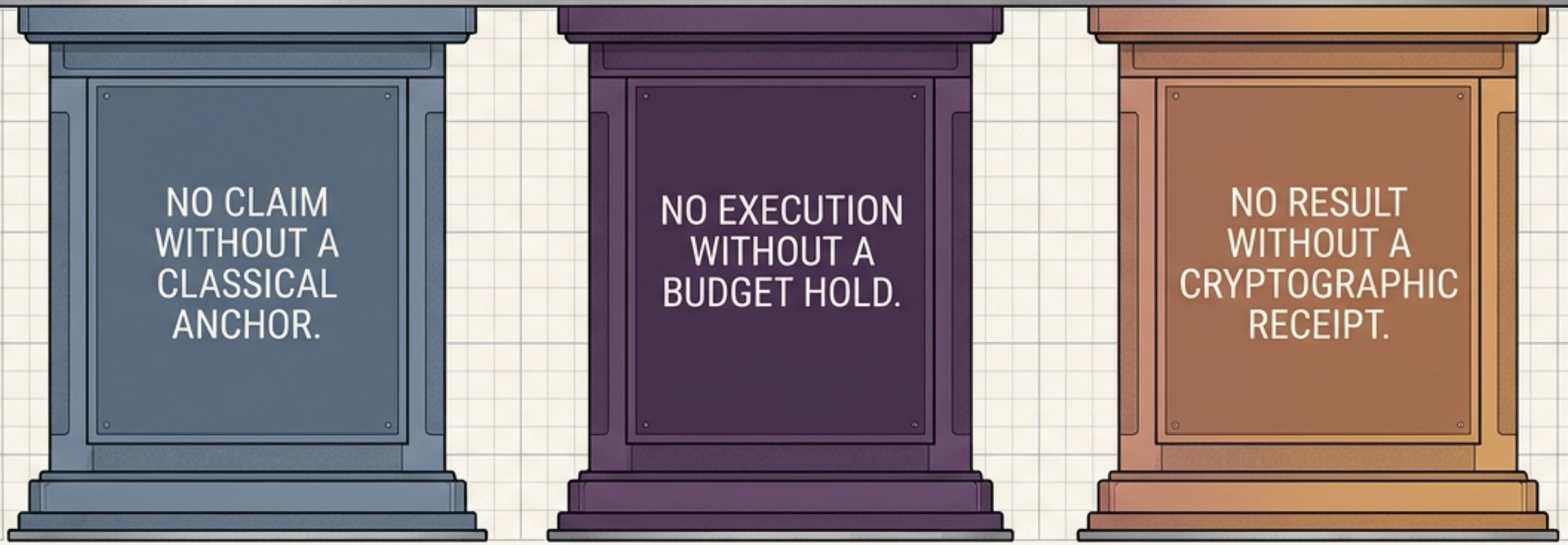
See, observe, discover — then know.



An open-source, local-first platform for autonomous AI agents conducting quantum computing and scientific discovery.

Gemini Notebook

EPISTEMIC GOVERNANCE & DISCOVERY



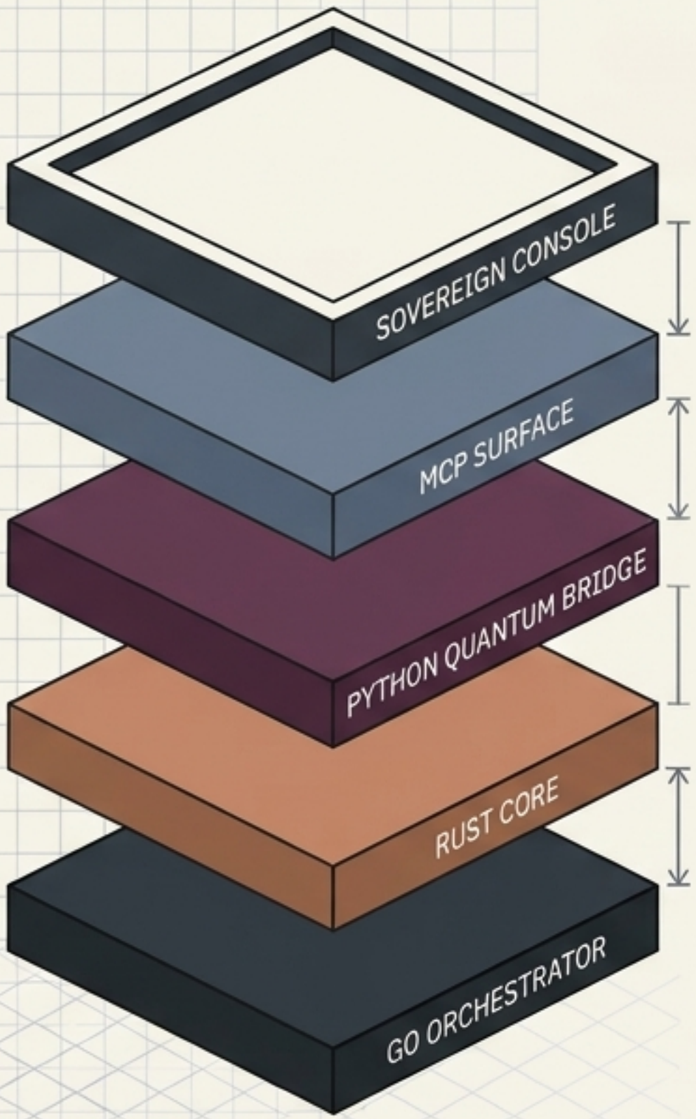
Machine-enforced taxonomy strictly grades all quantum outcomes against live classical baselines (never claiming advantage by assertion).

Pre-execution mandates and fail-closed gates prevent runaway cloud costs across quantum hardware.

Immutable provenance for every experiment, sealed into an Ed25519-signed WAL ledger.

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SYSTEM TOPOLOGY: THE MACRO MAP



SOVEREIGN CONSOLE

Next.js operator workbench with 3D Spatial Canvas.

`:3000`

MCP SURFACE

6 servers and 38 tools exposing state and execution to AI agents.

`Unix Domain Sockets`

PYTHON QUANTUM BRIDGE

Compute service bridging PySCF, Qiskit Aer, and PennyLane over a Unix Domain Socket.

RUST CORE

Compiler featuring `qir-core` (OpenQASM 3.0 to LLVM IR lowering) and `ifc-core` (information-flow confidentiality).

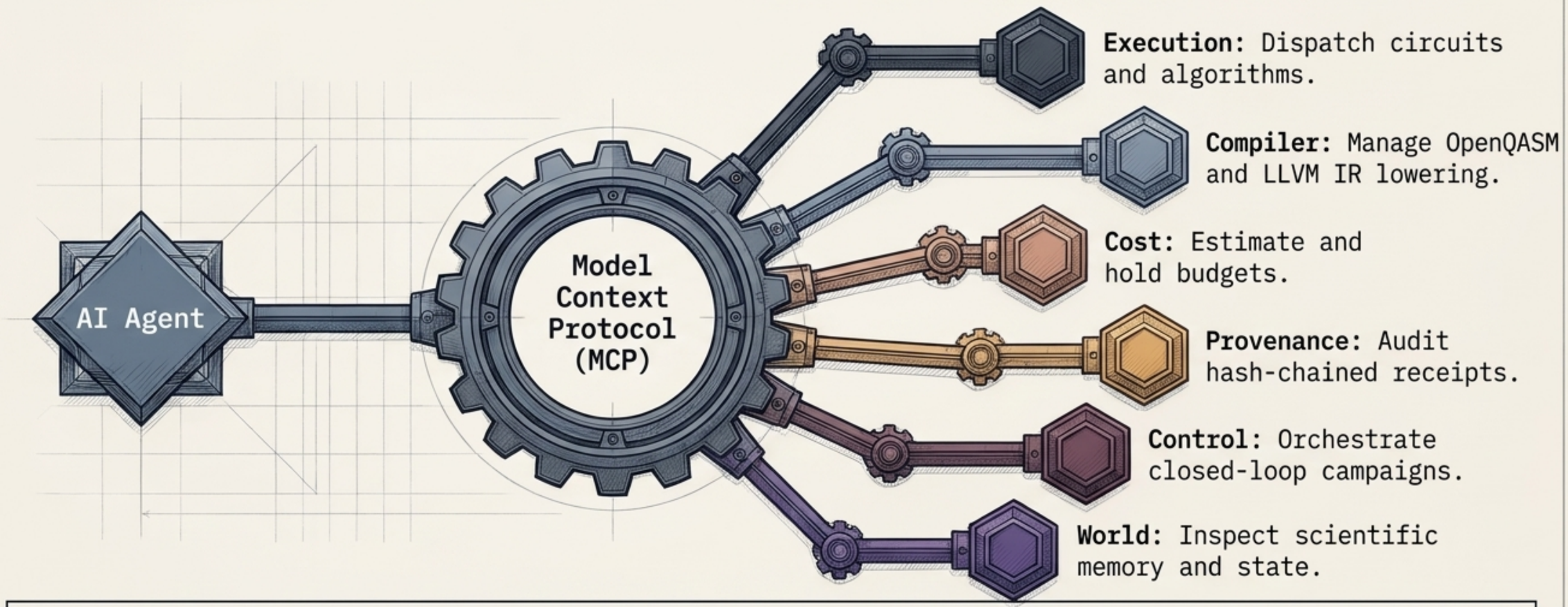
`:8080`

GO ORCHESTRATOR

The sovereign brain. Handles the HTTP/SSE API, hash-chained ledger, job admission, and rootless Docker/runsc sandbox execution.

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The Autonomous AI Interface (MCP Layer)



38 distinct tools over stdio allow autonomous agents to plan campaigns, inspect state, and audit cryptographic receipts natively.

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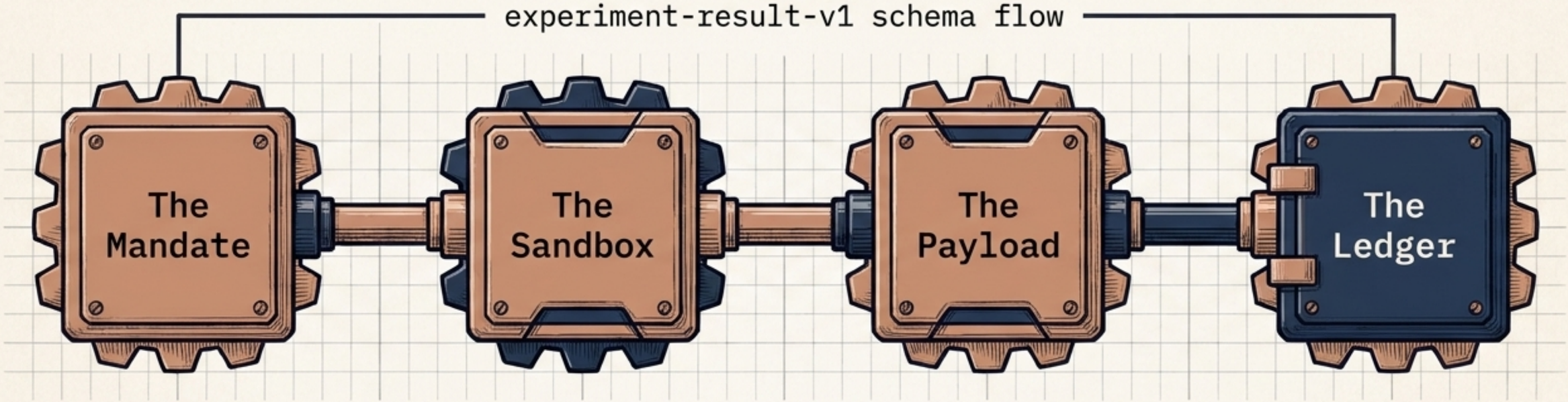
The Quantum & Chemistry Bridge



The Bridge allows seamless transit of ansatz transfer priors from classical optimization directly into quantum circuits.

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Cryptographic Provenance & Spend Settlement



Operator-signed WebCrypto Ed25519 mandate ensures pre-execution budget hold.

Secure runtime inside a rootless Docker/runsc execution environment.

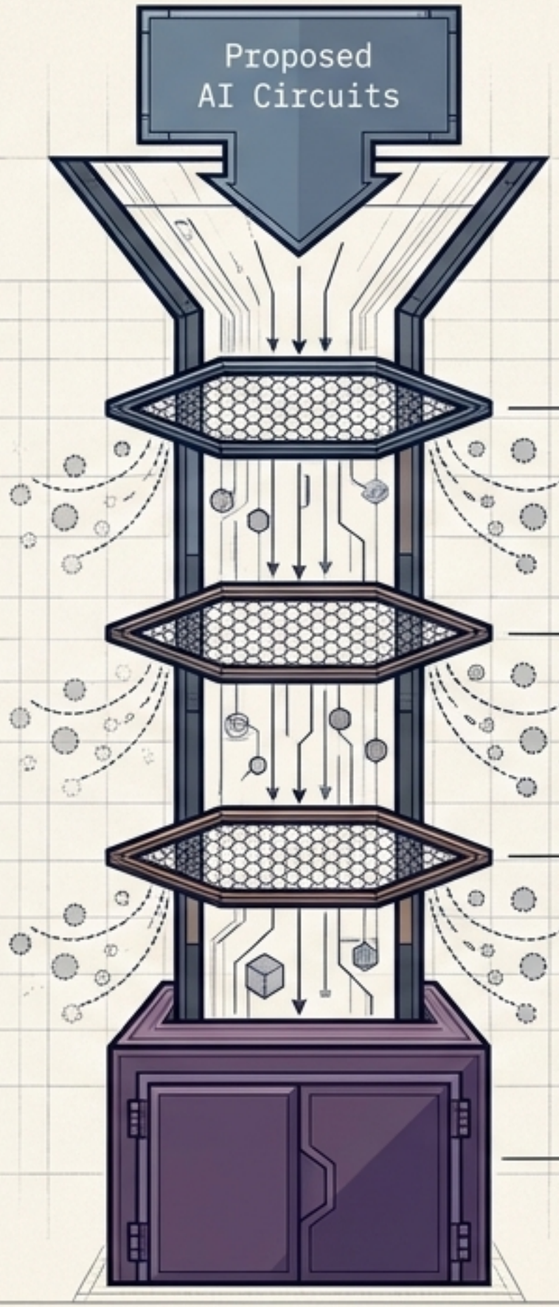
Generation of a strict, schema-validated JSON payload.

Sealed permanently into a SHA-256 hash-chained SQLite Write-Ahead Log (WAL).

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Quality Gates & Pre-Dispatch Fidelity

Protecting valuable QPU time and budget from noisy, useless executions.



Gate 1: Hardware T1 / T2 Evaluation
(Validates coherence times against circuit depth).

Gate 2: Two-Qubit Gate Error Thresholding
(Blocks circuits exceeding acceptable noise limits).

Gate 3: Calibration Freshness
(Requires hardware calibration profiles to be under 24 hours old).

Destination Floor:
Admitted high-fidelity circuits dynamically routed to IBM Quantum, IonQ, AWS Braket, or Local Simulators.

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The Governed Experiment Matrix

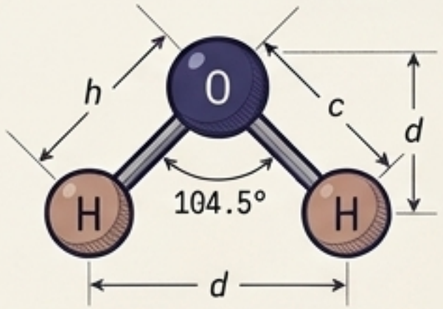
Capability Kind	Quantum Measurement	Classical Anchor
Molecular VQE	Shot-sampled Pauli expectation energy	Live PySCF RHF / CASCI / FCI
1D Spin Chain	Variational energy, magnetization, gap	Exact diagonalization (COBYLA / L-BFGS-B)
MaxCut QAOA	Shot-sampled cut value & approx ratio	Exact brute-force MaxCut
Math Conjecture	Symbolic/rational search	Finite-scope holdout certificate

Kubona never claims quantum advantage by assertion.
Every run is automatically graded against its classical anchor
(Verified, Supported, Plausible, Speculative, or Refuted).

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Feature Spotlight: Physics & Many-Body Simulation

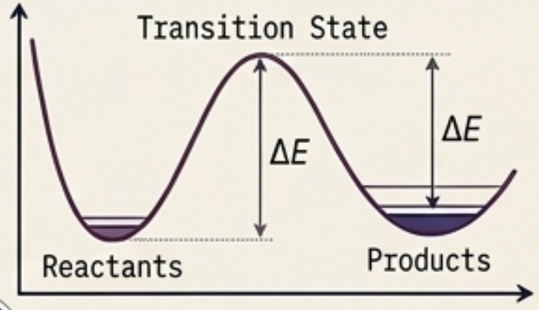
Chemistry & Geometry



Ab Initio & Variational Chemistry (VQE).

Bond-distance PES scans & BFGS equilibrium geometry optimization.

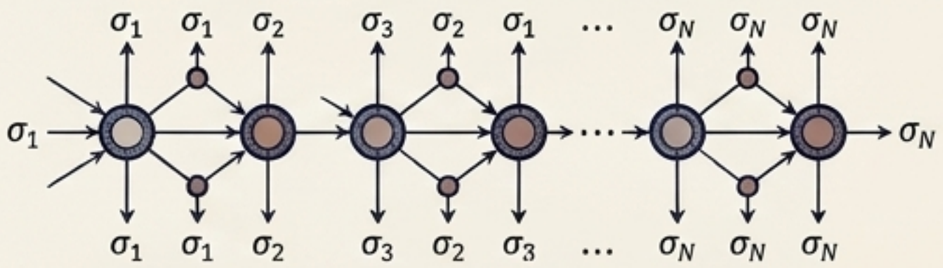
Reaction Energy Dynamics



Reactant-to-product ΔE calculations for H_2, LiH, BeH_2, H_2O.

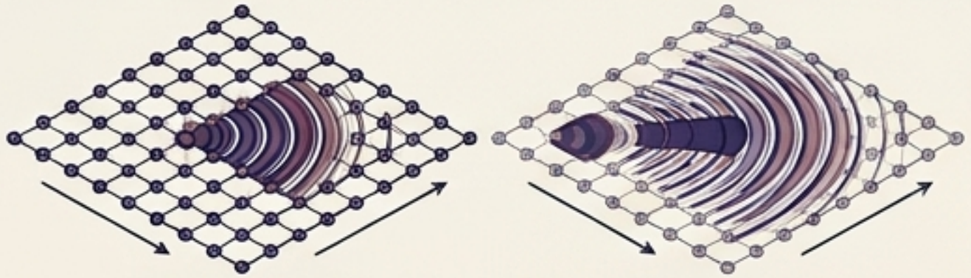
Ground + overlap-de-flated excited states.

Condensed-Matter & Tensor Networks



MPS / DMRG finite-size scaling to the thermodynamic limit ($1/N \to 0$).

2D Lattice Quench Dynamics



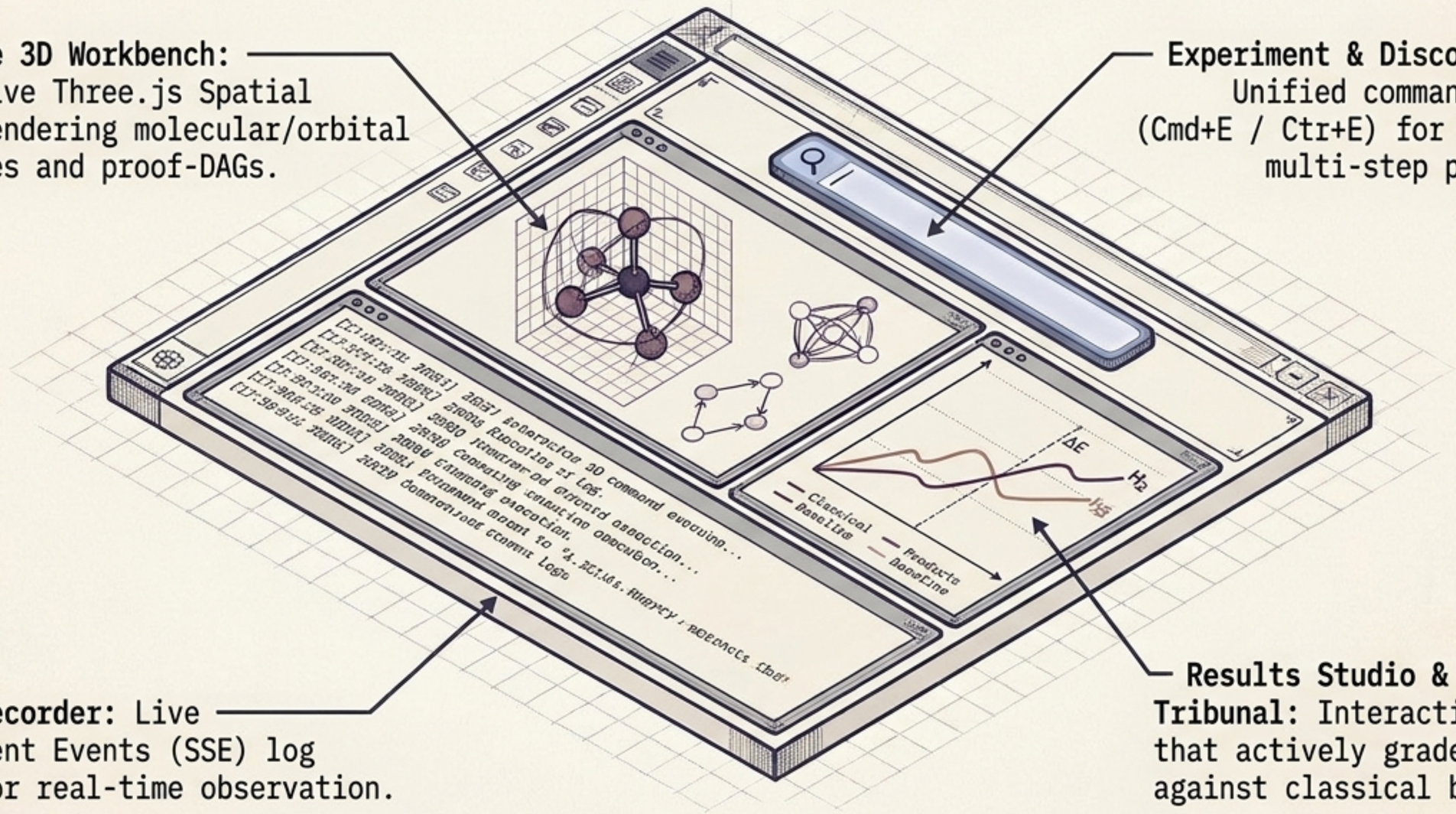
Exact finite-size 2D TFIM exact trajectories.
Sparse Krylov expm_multiply time evolution.

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The Sovereign Observatory (Console)

Real-Time 3D Workbench: Interactive Three.js Spatial Canvas rendering molecular/orbital geometries and proof-DAGs.

Experiment & Discovery Lab: Unified command palette (Cmd+E / Ctr+E) for launching multi-step protocols.



Flight Recorder: Live Server-Sent Events (SSE) log stream for real-time observation.

Results Studio & Epistemic Tribunal: Interactive charts that actively grade outcomes against classical baselines.

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Deployment Profile: Local-First Production

Local-Private Production

- Engineered for single-host (127.0.0.1 bindings for API and Console).
- All local simulators, classical anchors, and local LLMs run entirely offline.
- Cost: $0.00. Fail-closed budget gates ensure no accidental spend.

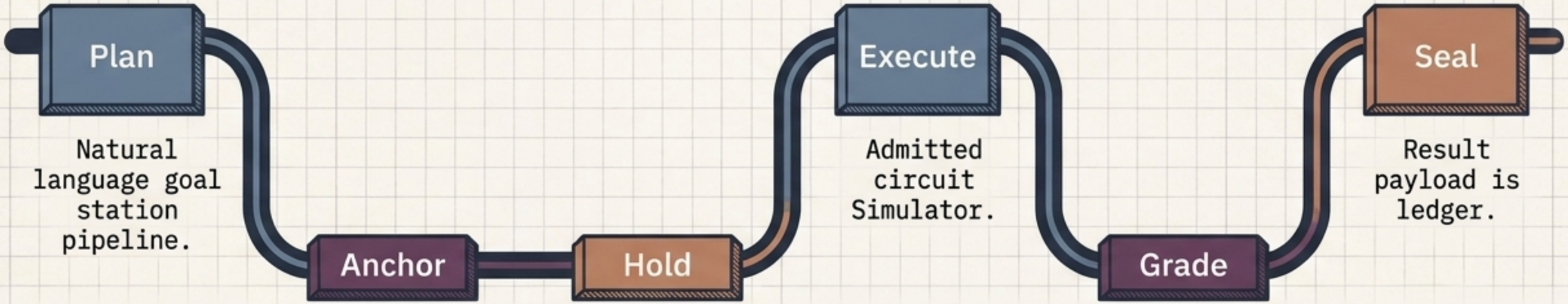


Cloud QPU (Explicit Opt-In)

- Strictly gated by operator-signed Ed25519 mandates.
- Cloud LLMs (Bring Your Own Key) and dispatch to IBM/AWS hardware require explicit enablement.

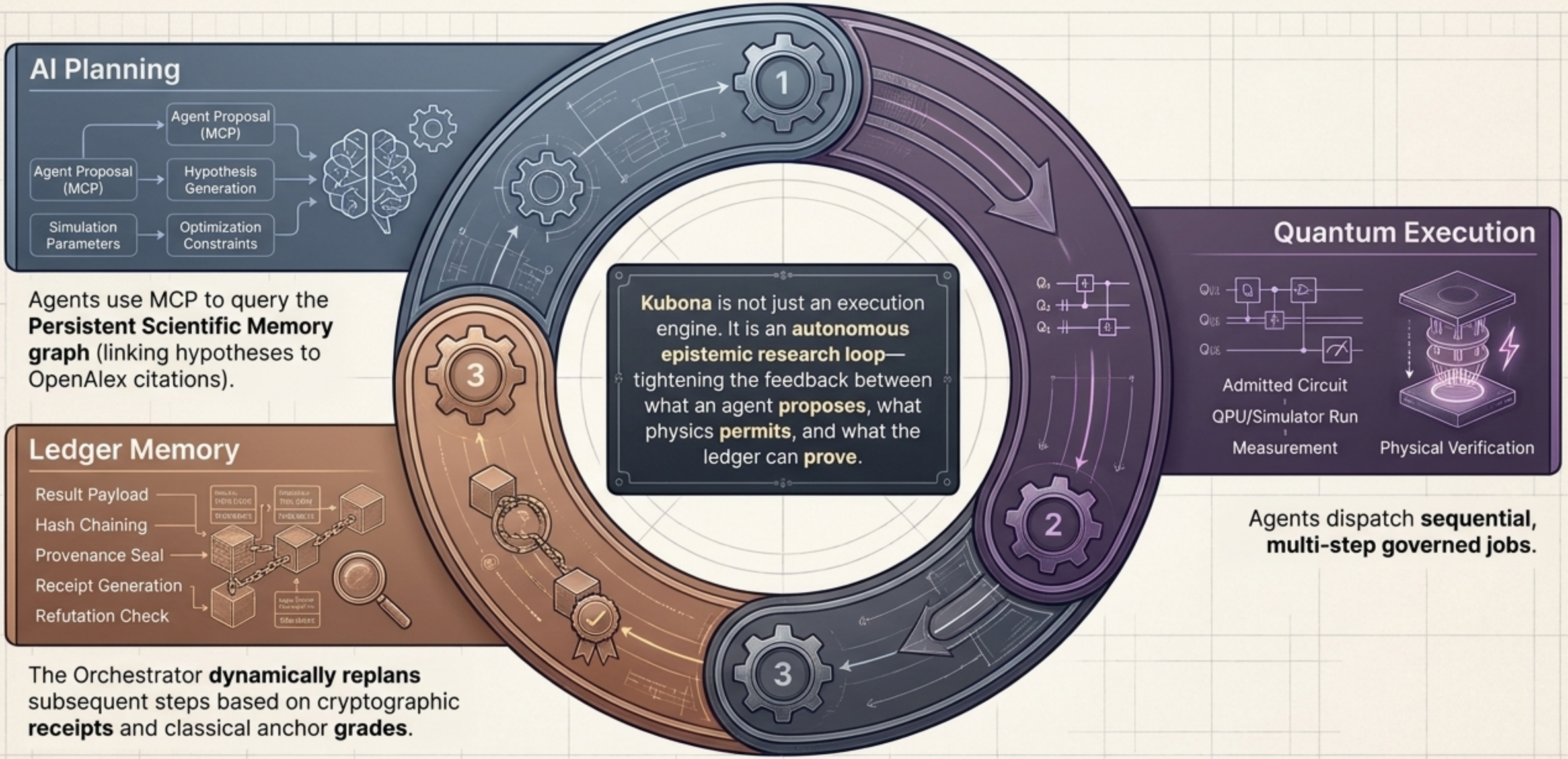
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Process Flow: The Life of an Experiment



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Synthesis: The Closed-Loop Campaign



Agents use MCP to query the **Persistent Scientific Memory graph** (linking hypotheses to OpenAlex citations).

Kubona is not just an execution engine. It is an **autonomous epistemic research loop**—tightening the feedback between what an agent **proposes**, what physics **permits**, and what the ledger can **prove**.

Agents dispatch **sequential, multi-step governed jobs.**

The Orchestrator **dynamically replans** subsequent steps based on cryptographic **receipts** and classical anchor **grades**.

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Quickstart & Developer Ecosystem

The Docker Stack

```
docker compose ... up --build -d
```

Note: Spins up the bridge, preflight gates, API, and console in one command.

The API Call

```
curl -X POST http://127.0.0.1:8080/api/experiments \
  -d '{"session_id":\"<session-id>",
"spec":"{"kind\":"spin_chain",\"model\":"xxz","anisotropy\":1.5},\"budget_usd":1}'
```

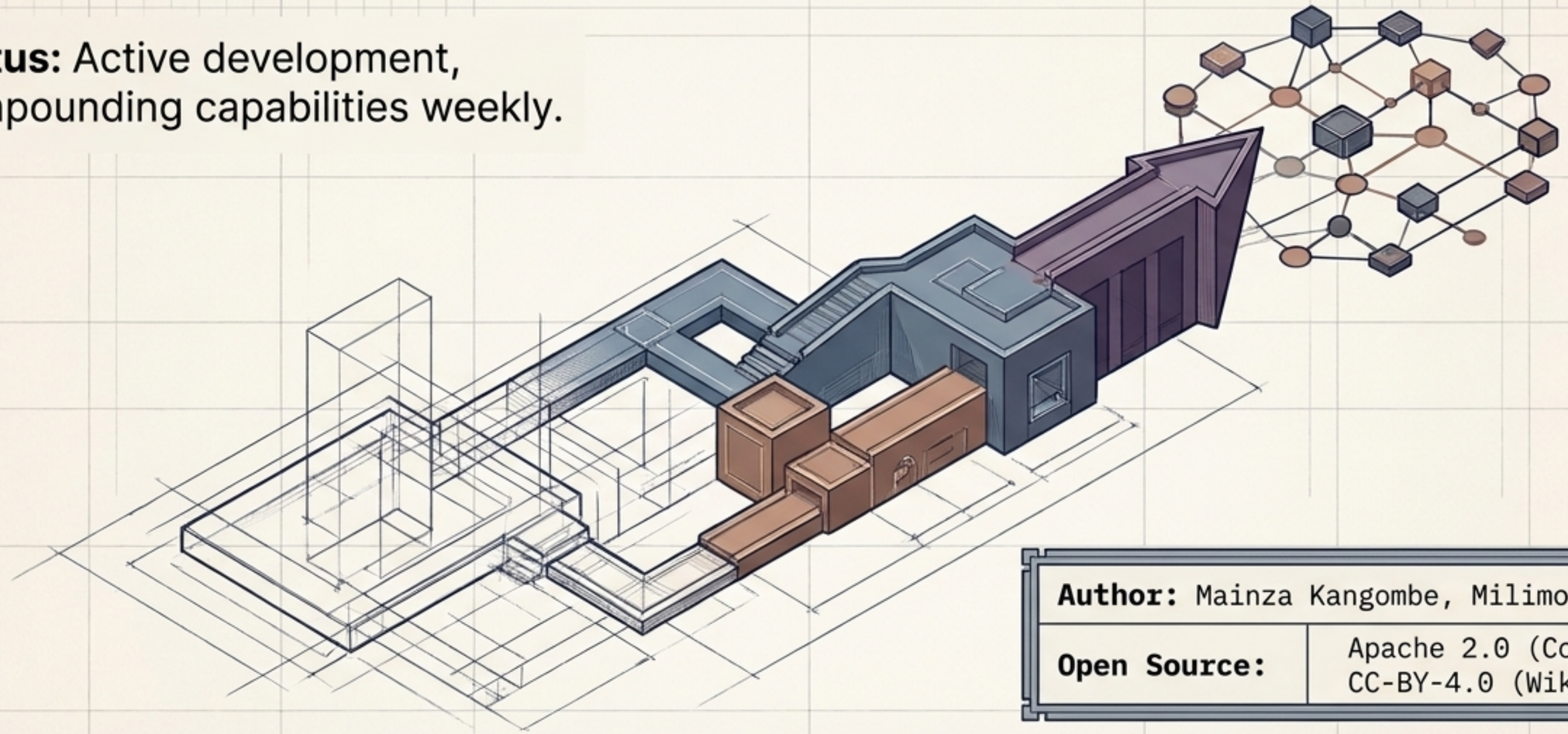
Verification Gates

`go test` `cargo test` `pytest` `eslint` `scripts/wiki-lint.py --strict`

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Roadmap & The Open-Source Call

Status: Active development, compounding capabilities weekly.



Author: Mainza Kangombe, Milimo Quantum.	
Open Source:	Apache 2.0 (Code) | CC-BY-4.0 (Wiki/Docs).

Clone the repository, run the deployment doctor (kubona.run doctor), and join the epistemic frontier at **github.com/mainza-ai/kubona**

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